Prime-supported Pell growth and binomial sequence spacing

An explicit quartic in 209 unrestricted integers whose unsolvability is equivalent to Con(ZFC)

Research prepared for Vladimir Reshetnikov and ProveIt
AI-assisted construction, proof, and independent arithmetic review

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Abstract

We give a fully specified polynomial $P \in \mathbb{Z}[x_0, \ldots, x_{208}]$ of degree four and an ordinary proof, using the explicitly identified inherited calibration of the numerical calculus $\mathcal{C}_{222}$, that ZFC proves $\neg\exists \boldsymbol{x} \in \mathbb{Z}^{209}\, P(\boldsymbol{x}) = 0$ equivalent to Con(ZFC). The polynomial is a sum of 150 squares of quadratic residuals; all residuals are printed in this report. Its collected expansion has 5,283 nonzero monomials and coefficient height 274,899,927,541.

The new construction couples two previously separate witnesses. A Pell ordinate supplies both a large growth scale and divisibility by every small prime. An already present binomial selector then supplies the capacity needed for the spacing of a Gödel sequence code. This removes the independent spacing multiplier. The key completeness argument chooses the proof horizon from the cofinal family $t = 32c^{2k}$, with the fixed prime $c = 524309$; finite-field bounds on Pell annihilating indices make the necessary congruences compatible.

Relative to the frozen incoming 210-variable quartic, the construction saves one unrestricted integer variable and reduces coefficient height by almost a factor of four, while adding one collected monomial. It does not establish a globally simplest equation or worldwide priority. The unbounded proof is distinct from the supplied exact coefficient checks. No new end-to-end Lean/Rocq verification or proof of consistency is claimed.

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1 Result, comparison, and meaning of simplicity

The primary result is the concrete equation

$$\boxed{P(x_0,\dots,x_{208}) := \sum_{j=0}^{149} R_j(x_0,\dots,x_{208})^2 = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^{209}.} \tag{1}$$

Appendix A specifies every R_j as a literal polynomial with integer coefficients. These 209 coordinates are all unrestricted integers and all occur in the collected polynomial. Auxiliary letters used in the mathematical explanation are abbreviations or proof parameters; they do not enlarge the variable list in (1).

Theorem 1.1 (Main result)**.** *For the literal polynomial in Appendix A, using the retained ordinary arithmetic and $\mathcal{C}_{222}$/ZF/ZFC calibration described in Section 3, there is a proof formalizable in ZFC of*

$$\big(\exists \boldsymbol{x} \in \mathbb{Z}^{209}\ P(\boldsymbol{x}) = 0\big) \quad \longleftrightarrow \quad \exists p\ \mathrm{Proof}_{\mathsf{ZFC}}(p, \bot).$$

Consequently,

$$\mathsf{ZFC} \ \vdash\ \big(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{209}\ P(\boldsymbol{x}) = 0\big) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

Here $\mathrm{Proof}_{\mathsf{ZFC}}$ refers to the fixed effective presentation used by the inherited calibration. The assertion is internal: the proof codes and integers are those quantified over in ZFC. No restriction to externally standard codes, transitive models, or externally well-founded models is added. The equivalence does not assume consistency or inconsistency, and does not decide which side holds.

The immediate comparison is the incoming package `ConZFC_Pell_Growth_210_2026-10-25.zip` at the pinned repository commit [2, 1]. Its complete bytes are retained in this release. Both columns below were independently collected from their literal residual lists.

Measure	Frozen parent	This report
Unrestricted integer variables	210	**209**
Quadratic residual squares	150	150
Exact total degree	4	4
Collected monomials	5,282	5,283
Collected coefficient height	1,099,511,627,776	274,899,927,541
Largest residual coefficient	1,048,576	524,309
Residual monomials before squaring	999	999

The new height is about 25.0020% of the old one. The single extra collected monomial is an explicit cost. We report the expanded coefficient height, not just the largest scalar in a circuit.

There is no single established ordering called "simplest" that combines variable count, degree, coefficient height, support, circuit size, domain, and proof complexity. The inspected repository already contains equations with many fewer variables at other degrees and coefficient costs, including the incoming fourteen-positive-variable, 65-operation carrier equation [5]. This work improves the direct quartic family. It uses the numerical proof calculus, without a K215 simulation or an instantiation of Jones's universal polynomial. It does not show that 209 is a minimum, or that degree four is necessary for this particular consistency statement.

2 The new idea in arithmetic terms

The direct compiler needs two kinds of size. First, F must be large enough to separate products of formula weights $1 + Fj$. Second, a spacing y must be large enough to store the intermediate pending-multiset products in a Gödel β code. For completeness the moduli

$$1 + y,\ 1 + 2y,\ \ldots,\ 1 + (t + 1)y$$

must be pairwise coprime. A sufficient condition is that every prime at most t divide y.

The parent uses a freely chosen odd multiplier:

$$y = F(2v + 1).$$

It chooses v sufficiently large and with appropriate divisibility. We replace this by

$$\boxed{y = FB, \qquad B = \binom{F + t}{t}.} \tag{2}$$

The coordinate B and its exact polynomial authentication already exist. The new equation for y is quadratic and the coordinate v disappears. The binomial notation occurs in the proof, not as a primitive operation in the final polynomial.

This substitution has two independent obligations. The value FB must exceed every state of the selected finite proof. Also FB must have enough small prime divisors for the beta moduli. The binomial supplies the first property through its degree in F. We arrange the second directly in F.

Put

$$d = 2^{19} = 524288, \qquad c = 524309. \tag{3}$$

The constant c is prime: exact trial division by every prime not exceeding $\lfloor \sqrt{c} \rfloor = 724$ verifies this finite fact. Impose

$$t = 32(2z + 1)^2, \tag{4}$$

$$Z = ctF, \tag{5}$$

$$X^2 - Z^2 - 2ZF = 1, \tag{6}$$

$$F + t = ctu. \tag{7}$$

All four residuals have degree at most two. The middle pair gives

$$X^2 - \big((ct + 1)^2 - 1\big)F^2 = 1. \tag{8}$$

The congruence $F \equiv -t \pmod{ct}$ will force a sufficiently large positive Pell index, once the existing signed arithmetic front has established $F > 0$.

For completeness we may select horizons from the particular cofinal family

$$t = 32c^{2k}, \qquad z = (c^k - 1)/2, \qquad k \geq 1. \tag{9}$$

An arbitrary zero of the final polynomial need not have this special form. It only satisfies (4). The stronger choice is made when constructing a zero from a finite refutation. This distinction is essential.

The role of the prime c is structural. The modulus ct contains one more factor of c than t does, but every small-prime Pell annihilating index has fewer than $2k + 1$ factors of c. Therefore its required divisibility can coexist with the residue $n \equiv -t \pmod{ct}$. This is the new compatibility argument that lets the growth witness absorb the prime support formerly delegated to v.

3 What is inherited and what is proved here

The retained parent package contains the literal polynomial source, exact integer projections, arithmetic proofs, the relevant calibration sources, and its own earlier source archive. It is cited as a proved predecessor, with the following boundaries made explicit.

3.1 Arithmetic interfaces retained from the predecessor

The unchanged rows provide the following facts and constructions [2, 3, 4].

1. A shared signed Pell frame authenticates the retained power requests. The principal pairs are oriented before the signs of the arithmetic front are inferred. At any authenticated base of absolute value at least two, an integral power output forces its nominal exponent to be nonnegative and equals the intended power.
2. The actual CRT index representative C is even and nonzero. Its Pell pair is the reused anchor. The cap bounds all other retained nominal indices strictly below C after nonnegativity is recovered.
3. The arithmetic front retains

$$B\eta = C + F + 1, \qquad D = 2(C + F + 1), \qquad q = D^x, \tag{10}$$

$$x = F + 1 + (y + 1)r_0, \qquad x = 1 + \big(1 + (t + 1)y\big)r_t. \tag{11}$$

   The powers with base q and exponent $y + 1$, and with base $q + 1$ and exponents C and $F + t$, are among the retained requests.
4. Exact digit extraction authenticates $B = \binom{F+t}{t}$ and the range-binomial predicates. The shared remainder bound and the affine radix D suffice because all relevant upper indices are at most C and $q = D^x > (2C)^x$.
5. The factorial-free binomial CRT construction encodes the local calculus constraints. Its completeness requires the established growth bound on F, an odd selector B, a suitable positive beta spacing and cofinal choices of the CRT representatives.

We do not delete or weaken an anchor, orientation, cap, range, or remainder row. The new X, Z, u are private growth coordinates; they are not new requested atoms of the shared power frame. The old actual front atoms $F, B, t, y + 1$ are already present in that frame.

3.2 Logical calibration

The direct arithmetic source uses an inherited square-shell recoding of the numerical calculus $\mathcal{C}_{222}$. Its syntax translation fixes truth code 0 and falsity code 1 and transports every guarded rule instance. The displayed square-shell constructors are not literally the original Lean constructors. Compose this recoding with the retained original-calculus interfaces `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation` [3]. Their ordinary proofs supply the connection with the fixed ZFC presentation. The first interface is a statement about the numerical calculus itself; the occurrence of "KEndpoint" in a file or namespace does not insert a Turing-machine simulation into this construction.

The source includes a depth certificate for the square-shell constructor trees, and a Horn-flow proof extracting a finite refutation from the authenticated local arithmetic. These are inherited mathematical inputs. This report proves the new growth, prime-support, capacity, sign, and completeness arguments in full. It does not reprove the entire logical calibration or package a complete fresh Lean dependency closure.

Thus the result is an ordinary mathematical proof using named predecessor lemmas, together with exact finite verification of the delivered polynomial. It is not a newly checked object-language ZFC derivation. The successful Python checks are not a substitute for the unbounded arithmetic or logical theorems.

4 Elementary Pell facts

For an integer $a \geq 2$, define

$$X_0(a) = 1, \quad Y_0(a) = 0,$$

and

$$\begin{aligned} X_{n+1}(a) &= aX_n(a) + (a^2-1)Y_n(a), \\ Y_{n+1}(a) &= X_n(a) + aY_n(a). \end{aligned}$$

Equivalently,

$$X_n(a) + Y_n(a)\sqrt{a^2-1} = \left(a + \sqrt{a^2-1}\right)^n.$$

The recurrence, rather than real exponentiation, suffices for all subsequent formalization claims.

Lemma 4.1 (Classification and elementary bounds)**.** *For $a \geq 2$, every integral solution of $X^2 - (a^2-1)Y^2 = 1$ with $Y \neq 0$ has $|X| = X_n(a)$ and $|Y| = Y_n(a)$ for some positive integer n. Moreover,*

$$Y_n(a) \equiv n \pmod{a-1}, \tag{12}$$

$$Y_n(a) \geq (2a-1)^{n-1} \quad (n \geq 1). \tag{13}$$

If a is odd, then

$$\nu_2(Y_n(a)) = \nu_2(n) \quad (n \geq 1). \tag{14}$$

Proof. For classification take $x = |X|$, $y = |Y| > 0$. If $y = 1$, then $x = a$. In general $(a-1)y < x \leq ay$, and multiplying $x + y\sqrt{a^2-1}$ by $a - \sqrt{a^2-1}$ gives another nonnegative integral norm-one pair with ordinate $ay - x$ strictly smaller than y. Descent reaches $(1, 0)$. Reversing the descent proves classification and uniqueness of the positive index.

The recurrence defines integer polynomials in a. At $a = 1$ it gives $X_n(1) = 1$ and $Y_n(1) = n$, proving (12). The second-order recurrence $Y_{n+1} = 2aY_n - Y_{n-1}$ and strict increase imply $Y_{n+1} \geq (2a-1)Y_n$; use $Y_1 = 1$ for (13).

When a is odd, the recurrence modulo two makes every X_n odd and $Y_n \equiv n \pmod 2$. The doubling identity $Y_{2m} = 2X_mY_m$ then proves (14) by writing $n = 2^r s$ with s odd. □

Lemma 4.2 (A bounded annihilating index)**.** *Let $a \geq 2$, $\delta = a^2 - 1$, and let p be an odd prime. Set*

$$\rho_p = \begin{cases} p, & p \mid \delta, \\ p - \left(\dfrac{\delta}{p}\right), & p \nmid \delta, \end{cases}$$

where the second line uses the Legendre symbol. Then $1 \leq \rho_p \leq p+1$ and $p \mid Y_{\rho_p}(a)$. Any positive multiple of ρ_p has the same vanishing property. If a is odd, we may take $\rho_2 = 2$.

Proof. If $p \mid \delta$, the recurrence modulo p gives $Y_n(a) \equiv na^{n-1}$ and $a \equiv \pm 1$, so $n = p$ works. Otherwise let $s^2 = \delta$ in $\mathbb{F}_p$ when δ is a square, or in $\mathbb{F}_{p^2}$ when it is a nonsquare. Put $w = a + s$; its inverse is $a - s$. In the split case $w^{p-1} = 1$. In the nonsplit case Frobenius sends s to $-s$, so $w^p = w^{-1}$ and $w^{p+1} = 1$. In either case

$$Y_n(a) = \frac{w^n - w^{-n}}{2s}$$

vanishes at the stated index. The denominator is nonzero. The addition identities, or powers of w, give vanishing at all multiples. For $p = 2$ and odd a, use $Y_2(a) = 2a$ and the addition identities. □

The lemma does not assert that ρ_p is the smallest possible index. Only its explicit upper bound is needed; no theorem about minimal ranks of apparition is required.

5 Growth and the signed arithmetic front

5.1 The inherited label bound

The square-shell constructors are

$$I(u,v) = 2\big((u+v)^2 + u\big), \qquad M(u,v) = I(u,v) + 1,$$
$$A(u,v) = 2\big((u+v)^2 + 3u + 4v + 3\big), \qquad E(u,v) = A(u,v) + 1.$$

For $0 \le u, v \le M_0$, each constructor value plus one is at most

$$E(M_0, M_0) + 1 = 8M_0^2 + 14M_0 + 8 \le 8(M_0 + 1)^2.$$

With each local parameter below t, a depth-r tree therefore has value plus one at most $8^{2^r-1}t^{2^r}$. The retained source certificate has maximum constructor depth 19. Hence its local labels satisfy

$$0 \le j < L, \qquad L = 8^{d-1}t^d, \qquad d = 2^{19}. \tag{15}$$

This is a bound on decoded local values. It is not an upper bound on their large CRT representatives.

5.2 A sign-free bound comes first

By (4), $t \ge 32$ and t is even. Equation (7) says $F = t(cu - 1)$, so $F \ne 0$ because $c \nmid 1$, and F is even. Apply Lemma 4.1 at $a = ct + 1$. There is $n \ge 1$ with $|F| = Y_n(a)$. The congruence (12) and $F \equiv -t \pmod{ct}$ imply

$$n \equiv t \text{ or } -t \pmod{ct}.$$

Both least positive residues are at least t. Thus

$$|F| \ge (2ct + 1)^{t-1} > t. \tag{16}$$

No assertion about the sign of F has been used.

The shared frame establishes its canonical-index conclusions independently of these new private coordinates. Since C and F are even, $C + F + 1$ is odd. The selector relation in (10) makes B and η odd and nonzero. Therefore the new relation $y = FB$ gives

$$|y| \ge |F| > t. \tag{17}$$

5.3 Recovering signs without a circular assumption

Here is the complete order needed from the retained power and endpoint interfaces.

1. $D = 2(C + F + 1)$ is even and nonzero, so $|D| \geq 2$. The request $q = D^x$ implies $x \geq 0$.
2. If $x = 0$, the second endpoint in (11) makes $1 + (t+1)y$ divide -1. But its absolute value exceeds one by (17). Hence $x \geq 1$ and $|q| \geq 2$.
3. The request at base q and exponent $y+1$ forces $y+1 \geq 0$. A negative y would have $y \leq -|F| < -t$, which is impossible. Thus $y \geq |F| > 0$.
4. If $x = 1$, the first endpoint gives $-F = (y+1)r_0$, contrary to $0 < |F| \leq y < y+1$. Hence $x \geq 2$. The second endpoint now yields $r_t \geq 1$ and
$$x \geq 2 + (t+1)y. \tag{18}$$
5. Consequently $|q| \geq 4$ and $|q+1| \geq 3$. The retained requests at base $q+1$ force their nominal indices nonnegative. In particular $F + t \geq 0$. Together with (16) this gives $F > 0$. The same argument gives $C \geq 0$; the anchor already excluded $C = 0$. Thus $C > 0$, $D > 0$, and $q > 0$.
6. Since $y = FB$ and $y, F > 0$, also $B > 0$. The first endpoint gives $r_0 \geq 0$ and $x \geq F + 1$.

Because B is odd, the old spacing multiplier could now be reconstructed as $v = (B-1)/2 \in \mathbb{Z}$ if needed to invoke a retained beta lemma. It is not a quantified coordinate of the new polynomial. All sign and remainder arguments used by the old arithmetic front therefore retain their premises.

5.4 Recovering the full separation budget

Now $F > 0$, so the ambiguous congruence resolves to

$$n \equiv -t \pmod{ct}, \qquad n \geq (c-1)t > dt.$$

Since $2c \geq 8$, Lemma 4.1 gives

$$F \geq (2ct+1)^{(c-1)t-1} \geq (8t)^{dt-1},$$
$$\frac{(8t)^{dt-1}}{L^t} = \frac{8^{t-1}}{t} \geq 1.$$

Therefore

$$F \geq L^t. \tag{19}$$

This is precisely the inherited product-separation hypothesis. It is derived after the sign bootstrap, not used to justify that bootstrap.

The unchanged affine-radix and digit arguments now authenticate the binomial selector and local range predicates. For example, $D > 2C$ and the cap imply that every required binomial coefficient is bounded by $C^x < q/2^x$. The selector relation also gives $|B| \leq D/2 < q/2$ before its digit is identified. The retained common remainder bounds select the exact digit. The inherited separation and Horn-flow arguments then recover a finite $\mathcal{C}_{222}$ refutation. This proves soundness of the modified source.

6 The prime-support completeness theorem

Theorem 6.1 (Cofinal Pell ordinates with small-prime support)**.** *Let $c = 524309$ and $t = 32c^{2k}$ with $k \geq 1$. Put $a = ct + 1$. For any integer $\ell \geq 1$, there are arbitrarily large positive integers n satisfying*

$$n \equiv -t \pmod{ct}, \qquad 2^\ell \mid Y_n(a), \qquad p \mid Y_n(a) \quad \text{for every prime } p \leq t.$$

Consequently the new growth equations have cofinally large positive solutions F with all those divisibility properties.

Proof. For each prime $p \leq t$ choose the explicit ρ_p of Lemma 4.2, including $\rho_2 = 2$. Set

$$M = \operatorname{lcm}(2^\ell, \{\rho_p : p \leq t\}).$$

Since $c > 33$,

$$\rho_p \leq p + 1 \leq t + 1 = 32c^{2k} + 1 < c^{2k+1}.$$

Thus $\nu_c(\rho_p) \leq 2k$, and $\nu_c(M) \leq 2k$. The only prime factors of $ct = 2^5 c^{2k+1}$ are 2 and c. Its intersection with M has 2-adic valuation at most five and c-adic valuation at most $2k$. Therefore

$$\gcd(M, ct) \mid 2^5 c^{2k} = t. \tag{20}$$

The two congruences

$$n \equiv 0 \pmod{M}, \qquad n \equiv -t \pmod{ct}$$

are compatible by the generalized Chinese remainder theorem. Their solutions form an arithmetic progression with positive period, so positive solutions are cofinal.

For such n, every ρ_p divides n; Lemma 4.2 gives $p \mid Y_n(a)$. Also a is odd and $2^\ell \mid n$, so Lemma 4.1 gives $2^\ell \mid Y_n(a)$. Put

$$F = Y_n(a), \quad X = X_n(a), \quad Z = ctF, \quad u = \frac{F + t}{ct}.$$

The last quotient is integral by (12); all three growth rows hold. Strict increase of $Y_n(a)$ makes the resulting positive ordinates cofinal as well. □

Corollary 6.2 (Odd selector and coprime beta moduli)**.** *Choose ℓ with $2^\ell > t$, take a positive F from Theorem 6.1, and define*

$$B = \binom{F + t}{t}, \qquad y = FB.$$

Then B is odd and the integers $1 + iy$, $1 \leq i \leq t + 1$, are pairwise coprime.

Proof. For $1 \leq i \leq t < 2^\ell$, divisibility $2^\ell \mid F$ implies $\nu_2(F + i) = \nu_2(i)$. Subtracting valuations in

$$B = \frac{\prod_{i=1}^{t}(F + i)}{t!}$$

gives $\nu_2(B) = 0$.

Every prime at most t divides F, and hence y. If a prime p divided both $1 + iy$ and $1 + jy$ for $i < j$, it would be coprime to y and divide $j - i$. Thus $p \leq j - i \leq t$, contradicting $p \mid y$. □

The proof requires only prime support in y, not $t! \mid y$. It also does not require every solution of the new polynomial to have the small-prime divisibility properties. Those are choices in the completeness construction; soundness already follows from the retained exact arithmetic.

7 Why the binomial selector has enough capacity

The number of available proof rows can be chosen strictly larger than the number of genuine backward-expansion steps. Starting from one pending formula, each step replaces one formula by at most two premises. A proof with h such steps therefore has at most $h+1$ pending formulas at any time. The retained truth padding leaves the pending product unchanged [2]. Choose the horizon t with $t > h+1$ and above all finite local parameter requirements. The family (9) permits this.

Lemma 7.1 (Binomial capacity)**.** *Let $t \geq 2$, $L \geq t$, and $F \geq L^t$ be integers. Suppose a state is a product of at most $t-1$ weights $1+Fj$, where $0 \leq j < L$. Then*

$$S < F\binom{F+t}{t}.$$

Proof. Since the labels are integers, $1+Fj \leq 1+F(L-1) \leq FL$. Thus $S \leq (FL)^{t-1}$, including the empty product. On the other hand,

$$F\binom{F+t}{t} \geq \frac{F^{t+1}}{t!}.$$

The ratio of the latter lower bound to the former upper bound is

$$\frac{F^2}{t!L^{t-1}} \geq \frac{L^{2t}}{L^tL^{t-1}} = L > 1,$$

where $t! \leq t^t \leq L^t$. This proves the strict inequality. □

Apply the lemma to (15) and (19). Every actual pending state is below $y = FB$, so it is a valid remainder for each relevant beta modulus. No additional growth request, inequality coordinate, or free spacing multiplier is needed.

There is also a useful alternative proof of the existence needed here. For any fixed proof and horizon, its pending states are polynomials in F of degree at most $t-1$, while $F\binom{F+t}{t}$ is a polynomial of degree $t+1$ with positive leading coefficient. The cofinal Pell progression eventually dominates them. The explicit estimate above is stronger: the already authenticated lower bound on F suffices.

8 Completeness and the internal consistency equivalence

Start with a finite $\mathcal{C}_{222}$ refutation. Unfold it to a finite backward proof schedule if necessary, and let h be its number of genuine expansion steps. This may enlarge the proof, but the construction has no imposed resource bound.

Choose k sufficiently large that $t = 32c^{2k}$ is greater than $h+1$ and all required finite local parameters. Then set $z = (c^k-1)/2$. Choose ℓ with $2^\ell > t$ and use Theorem 6.1 to construct n, F, X, Z, u. Define the already intended selector and spacing by

$$B = \binom{F+t}{t}, \qquad y = FB.$$

Corollary 6.2 makes the beta moduli pairwise coprime and B odd. Lemma 7.1 bounds every required pending state. Truth steps pad the proof to the selected horizon without enlarging those states.

Ordinary finite CRT now supplies a positive beta code x with the prescribed states and endpoints, and permits x to be arbitrarily large in its residue class. Set $N = B^x$. The unchanged factorial-free binomial CRT construction provides the residue classes for the ordinary representatives and C at precision N^2. The growth hypothesis is stronger than its lower-bound requirement.

Because N is odd, the prescribed class for C has arbitrarily large even positive representatives. Choose the other representatives first, then choose C even and larger than all other retained nominal indices, including $F + t$, x, and $y + 1$. The retained selector congruence gives $B \mid C + F + 1$. Set

$$\eta = (C + F + 1)/B, \qquad D = 2(C + F + 1), \qquad q = D^x.$$

The same binomial-digit bounds and $N \mid q$ properties as in the predecessor follow. Choose the common frame center and the canonical principal Pell pairs last. The inherited cap, orientation, power, remainder, and local arithmetic completeness constructions supply all remaining coordinates.

The order is explicit and acyclic: the finite proof fixes t; then the growth construction fixes F, B, y; then beta CRT fixes x and N; then the local CRT representatives and C are chosen; finally D, q and the shared frame are completed. No modulus is chosen using a coordinate that was required to have been fixed earlier.

Soundness was established in Section 5; completeness is the construction just given. Compose them with the retained $\mathcal{C}_{222} \leftrightarrow$ ZF refutation $\leftrightarrow$ ZFC refutation calibration to obtain Theorem 1.1. Every new step is finite arithmetic, induction, CRT, or elementary finite field algebra, and is formalizable in ZFC. The cofinal choices are proved existence claims inside that same theory.

9 From the semantic change to the literal polynomial

The retained parent equation has 210 coordinates and 150 residuals. Its exact SHA-256 is

`20a2108353a3756ed7d1ddfdc71695131dcf0b6d94039988d9565ac48e6e1f0a.`

The new equation is obtained by the following finite specification. All indices in this list are *parent* indices until the final renumbering.

1. Undo the parent's optimization $R_3 \leftarrow R_3 + R_{146}$ by replacing row 3 with $R_3 - R_{146}$. This is an invertible integer row operation and restores the literal selector relation before beta is changed.
2. Replace row 125 by $x_0 - 32 - 128x_{178} - 128x_{178}^2$.
3. Replace row 146 by $x_4 - 1 - x_2x_3$.
4. Replace rows 147–149 by

$$\begin{aligned} x_{208} - cx_0x_2, \\ x_{207}^2 - x_{208}^2 - 2x_2x_{208} - 1, \\ x_2 + x_0 - cx_0x_{209}. \end{aligned}$$

5. Delete the now-unused coordinate 206. Keep coordinates 0–205 unchanged and send old coordinates 207, 208, 209 to new coordinates 206, 207, 208.

No other residual is changed. Exactly the following semantic coordinates are relevant to the modification.

Quantity	Parent coordinate	Final coordinate
t	x_0	x_0
F	x_2	x_2
B	x_3	x_3
$y+1$	x_4	x_4
C	$2x_{16}$	$2x_{16}$
z	x_{178}	x_{178}
Old free multiplier	x_{206}	deleted
X	x_{207}	x_{206}
Z	x_{208}	x_{207}
u	x_{209}	x_{208}

The new equation JSON has SHA-256

`b0413e587c595c6813f8eb67ef1581a507cee6155a38b8d091ca064f3e29b9f6.`

The separate collected expansion fixes all 5,283 coefficients. In its term format, $[a, [i, j, k]]$ means $ax_ix_jx_k$; repeated indices mean powers and an empty list means a constant. The equation file fixes the residual squares instead, and is substantially easier to audit.

The semantic replacements are not asserted to be invertible maps on all parent zeros. In particular an arbitrary old refutation witness need not already have prime-supported F. The completeness proof constructs an appropriate new witness from the underlying finite refutation. Only the initial row operation and final unused-coordinate deletion are pure algebraic operations of the elementary kind stated above.

10 Verification, reproducibility, and scope

The release provides separate implementations for production and finite verification. The verifier does not import the producer's polynomial code or the predecessor's polynomial helpers. It authenticates the parent bytes, reconstructs the six changed rows, checks the renumbering and variable occurrence, independently collects every square, and compares the full expansion and metrics. It also verifies the fixed prime by trial division. Its checks remain active with Python optimization enabled.

Normal and `-O` runs pass. Seven deliberate residual corruptions and one collected-coefficient corruption are rejected. A separate parser checks every one of the 150 printed residuals and all 999 printed coefficients against the equation JSON, with a deliberate print-corruption control. The parent package's complete release verifier was freshly run; its 352-file manifest, inherited checks, source/projection chain, printed equations, and fresh reconstruction passed. The receipt records the actual scope rather than relabeling an old receipt as a new run.

The arithmetic regression program checks 1,248 local Pell annihilating cases, including 126 ramified cases. A smaller-prime completeness fixture uses $c = 37$, $t = 43,808$, 4,564 prime constraints, and a 10,420-bit index. It checks the congruences by modular powering rather than attempting to print the enormous Pell ordinate. These are finite debugging checks; the proof for all horizons is Theorem 6.1.

Two independently assigned AI review tasks checked the new mathematical arguments. Their notes are retained. This is not external human peer review. No numerical search for a root can validate the consistency equivalence, and none is presented as doing so.

From the extracted package root, Python 3.10 or later suffices for the finite checks:

```
python3 -B code/verify.py
python3 -B -O code/verify.py
python3 -B code/check_printed.py
python3 -B code/check_arithmetic.py
```

The literal equation can be regenerated with

```
python3 -B code/build.py
```

The evaluator accepts a JSON array containing exactly 209 integers:

```
python3 -B code/evaluate.py tuple.json --residuals
```

To rebuild the standalone article and PDF with a standard TeX Live installation, run

```
python3 -B code/build_report.py
```

PDF metadata or TeX versions can change PDF bytes. The finite equation identity checks are independent of such changes. `SHA256SUMS` authenticates the delivered files; it is not a theorem about the arithmetic.

11 What this adds, and the remaining research problem

The contribution is the proved reuse of a selector for sequence spacing, made possible by a cofinal Pell construction with controlled prime support. It couples two arithmetic tasks that were independent in the parent. The prime-supported ordinate supplies the coprimality requirement; the binomial selector supplies capacity; the existing front supplies sign recovery and the exact selector interpretation. None of these roles is left to an unspecified universal polynomial or an uncompiled variable exponent.

The quantitative gain is modest and exact: one fewer unrestricted integer unknown, unchanged degree and residual count, nearly four times smaller coefficient height, and one additional collected monomial. The argument supplies a reusable arithmetic lemma, which may have value beyond this one-coordinate saving. It does not establish that its proof complexity is smaller than the parent's.

Several possible shortcuts remain unproved and are not used. Taking $y = B$ loses the guaranteed small-prime support. Enforcing divisibility only by the common growth modulus does not make all beta moduli coprime. Choosing a generic horizon does not automatically solve the new index CRT compatibility problem. Dropping $F + t$ from the retained power requests would remove the stated sign-recovery argument. Treating the new source as pointwise equivalent to every old witness would also be unjustified.

The next substantive target is to apply the same prime-support principle to another already authenticated arithmetic quantity, so that another independent witness can be removed without raising degree. This report delivers the completed 209-variable construction; no further count is claimed without a literal polynomial and proof.

References

[1] Vladimir Reshetnikov, *ProveIt*, inspected snapshot `32e9e9eb5999c96b0efe80c80a4018afd33d44ae`. `https://github.com/VladimirReshetnikov/ProveIt/tree/32e9e9eb5999c96b0efe80c80a4018afd33d44ae`.

[2] *Pell growth from divisibility: an explicit quartic for Con(ZFC)*, incoming package `ConZFC_Pell_Growth_210_2026-10-25.zip`, same pinned snapshot. Relevant components are `proofs/growth_proof.md`, `proofs/AFFINE_RADIX_PROOF.md`, `equations/principal210/equation.json`, and the inherited anchor package. Exact archive retained in `inherited/`. `https://github.com/VladimirReshetnikov/ProveIt/blob/32e9e9eb5999c96b0efe80c80a4018afd33d44ae/docs/incoming/ConZFC_Pell_Growth_210_2026-10-25.zip`.

[3] *A reused CRT index: an explicit Diophantine consistency equation*, ProveIt source 173, `173-crt-index-anchor-215`, 25 October 2026. In particular `C_ANCHOR_THEOREM.md`, `LABEL_SEPARATION_WITHOUT_DERIVED_RANGE.md`, `CONSTRUCTOR_DEPTH.md`, and `ORDINARY_PROOF_DEPENDENCY_BOUNDARY.md`. Selected files and the relevant calibration sources are retained within the parent archive. `https://github.com/VladimirReshetnikov/ProveIt/tree/32e9e9eb5999c96b0efe80c80a4018afd33d44ae/Computability/DiophantineConsistency/con-zfc-diophantine-forms/173-crt-index-anchor-215`.

[4] *Linear CRT*, ProveIt source 157, `157-linear-crt`. The retained `MAIN_PROOF.md` supplies the factorial-free binomial CRT and signed arithmetic interfaces. See also the retained source-151 factorial-free report and source-158 pooled-power proofs within the parent archive. `https://github.com/VladimirReshetnikov/ProveIt/tree/32e9e9eb5999c96b0efe80c80a4018afd33d44ae/Computability/DiophantineConsistency/con-zfc-diophantine-forms/157-linear-crt`.

[5] *ConZFC Pell Carrier65*, incoming package `ConZFC_Pell_Carrier65_2026-10-25.zip`, same pinned snapshot. Used only to describe the broader comparison frontier, not as a logical dependency of the new quartic. `https://github.com/VladimirReshetnikov/ProveIt/blob/32e9e9eb5999c96b0efe80c80a4018afd33d44ae/docs/incoming/ConZFC_Pell_Carrier65_2026-10-25.zip`.

A The complete literal polynomial

All variables in the following list range independently over $\mathbb{Z}$. The polynomial is exactly $P = R_0^2 + \cdots + R_{149}^2$. Every residual has total degree at most two. This list is independently parsed back into integer coefficient dictionaries and compared with the machine-readable equation. The source contains no omitted rows, variable exponents, special function symbols, or externally supplied coefficients.

$$R_0 = x_0 - x_4 + x_7 - x_0x_4$$

$$R_1 = -x_2 - x_4x_5 + x_6x_7$$

$$R_2 = -1 + x_1 - x_6x_7$$

$$R_3 = 1 + x_2 + 2x_{16} - x_3x_{18}$$

$$\begin{aligned} R_4 = {} & -4 - 3x_8 - 4x_9 + x_{19} - x_8^2 \\ & - 2x_8x_9 - x_9^2 \end{aligned}$$

$$\begin{aligned} R_5 = {} & x_8 + x_{21} - x_8^2 - 8x_8x_{19} - 4x_8x_{20} \\ & - 16x_{19}^2 - 16x_{19}x_{20} - 4x_{20}^2 \end{aligned}$$

$$R_6 = 1 + 2x_{21} + x_{22} - 4x_{21}^2$$

$$R_7 = -2 + x_{23} - x_9^2 - 2x_9x_{10} - x_{10}^2$$

$$R_8 = 1 + x_{24} - x_8^2 - 2x_8x_{10} - x_{10}^2$$

$$\begin{aligned} R_9 = {} & -29 - 61x_8 + 16x_9 - 96x_{10} - 28x_{19} \\ & + 8x_{23} - 24x_{24} + x_{25} - 35x_8^2 - 96x_8x_{10} \\ & - 24x_8x_{19} - 4x_8x_{23} - 24x_8x_{24} - 2x_9^2 - 8x_9x_{23} \\ & - 64x_{10}^2 - 32x_{10}x_{19} - 32x_{10}x_{24} - 4x_{19}^2 - 8x_{19}x_{24} \\ & - 4x_{23}^2 - 4x_{24}^2 \end{aligned}$$

$$\begin{aligned} R_{10} = {} & -4 + 10x_8 + x_{26} - 2x_{43} - 2x_8^2 \\ & - 8x_8x_9 - 8x_8x_{23} - 4x_{25}^2 \end{aligned}$$

$$\begin{aligned} R_{11} = {} & -4 + x_{27} - 16x_{10}^2 - 16x_{10}x_{23} - 16x_{10}x_{24} \\ & - 4x_{23}^2 - 8x_{23}x_{24} - 4x_{24}^2 \end{aligned}$$

$$\begin{aligned} R_{12} = {} & -7 + 24x_{10} + 10x_{19} + 24x_{24} + 12x_{27} \\ & + x_{28} - 16x_{10}^2 - 16x_{10}x_{19} - 32x_{10}x_{24} - 16x_{10}x_{27} \\ & - 4x_{19}^2 - 16x_{19}x_{24} - 8x_{19}x_{27} - 16x_{24}^2 - 16x_{24}x_{27} \\ & - 4x_{27}^2 \end{aligned}$$

$$\begin{aligned}R_{13} = &-114-127x_8-67x_9-224x_{10}-16x_{23}\\&-40x_{24}+x_{29}-37x_8^2-24x_8x_9-112x_8x_{10}\\&-4x_8x_{23}-24x_8x_{24}-37x_9^2-112x_9x_{10}-24x_9x_{23}\\&-4x_9x_{24}-128x_{10}^2-32x_{10}x_{23}-32x_{10}x_{24}-4x_{23}^2\\&-4x_{24}^2\end{aligned}$$

$$\begin{aligned}R_{14} = &-30-14x_8-96x_9-128x_{10}-32x_{23}\\&+x_{30}-2x_8^2-24x_8x_9-32x_8x_{10}-8x_8x_{23}\\&-72x_9^2-192x_9x_{10}-48x_9x_{23}-128x_{10}^2-64x_{10}x_{23}\\&-8x_{23}^2-4x_{29}^2\end{aligned}$$

$$\begin{aligned}R_{15} = &-74-42x_8-84x_9+29x_{10}+21x_{19}\\&+40x_{23}+x_{31}-x_{59}+8x_8x_{10}+16x_8x_{23}\\&+16x_9x_{10}+32x_9x_{23}-3x_{10}^2-4x_{10}x_{19}-8x_{10}x_{23}\\&-8x_{19}x_{23}-4x_{23}^2\end{aligned}$$

$$\begin{aligned}R_{16} = &-131-184x_8-226x_{10}-88x_{23}-46x_{24}\\&-44x_{31}+x_{32}-64x_8^2-160x_8x_{10}-64x_8x_{23}\\&-32x_8x_{24}-32x_8x_{31}-100x_{10}^2-80x_{10}x_{23}-40x_{10}x_{24}\\&-40x_{10}x_{31}-16x_{23}^2-16x_{23}x_{24}-16x_{23}x_{31}-4x_{24}^2\\&-8x_{24}x_{31}-4x_{31}^2\end{aligned}$$

$$R_{17} = x_{20} - 4x_{19}^2$$

$$\begin{aligned}R_{18} = &\,x_8+x_{33}-x_8^2+8x_8x_{19}-4x_8x_{20}\\&-16x_{19}^2+16x_{19}x_{20}-4x_{20}^2\end{aligned}$$

$$R_{19} = 1 + 2x_{33} + x_{34} - 4x_{33}^2$$

$$R_{20} = x_{35} - x_9^2 - 4x_9x_{34} - 4x_{34}^2$$

$$\begin{aligned}R_{21} = &-44-52x_8-104x_9+50x_{19}+x_{36}\\&-16x_8^2-64x_8x_9+32x_8x_{19}-64x_9^2+64x_9x_{19}\\&-16x_{19}^2\end{aligned}$$

$$\begin{aligned}R_{22} = &\,8x_9+6x_{10}+4x_{36}+x_{37}-62x_9^2\\&-92x_9x_{10}-32x_9x_{23}-32x_9x_{36}-34x_{10}^2-24x_{10}x_{23}\\&-24x_{10}x_{36}-4x_{23}^2-8x_{23}x_{36}-4x_{36}^2\end{aligned}$$

$$\begin{aligned}R_{23} = &-90-152x_8-114x_{10}-38x_{24}-36x_{37}\\&+x_{38}-64x_8^2-96x_8x_{10}-32x_8x_{24}-32x_8x_{37}\\&-36x_{10}^2-24x_{10}x_{24}-24x_{10}x_{37}-4x_{24}^2-8x_{24}x_{37}\\&-4x_{37}^2\end{aligned}$$

$$R_{24} = x_{39} - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2$$

$$\begin{aligned} R_{25} = & -7 + 8x_8 + 6x_{24} + x_{40} - x_{43} \\ & - 3x_8^2 - 8x_8x_9 - 8x_8x_{23} - 8x_8x_{24} - 4x_8x_{39} \\ & - 8x_9x_{24} - 2x_{10}^2 - 8x_{10}x_{39} - 8x_{23}x_{24} - 4x_{24}^2 \\ & - 4x_{39}^2 \end{aligned}$$

$$\begin{aligned} R_{26} = & -4x_{10} - 2x_{24} - 4x_{39} + x_{41} - 16x_8^2 \\ & - 36x_8x_{10} - 32x_8x_{24} - 40x_8x_{39} - 16x_8x_{40} - 22x_{10}^2 \\ & - 32x_{10}x_{24} - 48x_{10}x_{39} - 16x_{10}x_{40} - 16x_{24}^2 - 32x_{24}x_{39} \\ & - 16x_{24}x_{40} - 24x_{39}^2 - 16x_{39}x_{40} - 4x_{40}^2 \end{aligned}$$

$$\begin{aligned} R_{27} = & x_{10} + x_{42} - 25x_{10}^2 - 40x_{10}x_{24} - 20x_{10}x_{27} \\ & - 16x_{24}^2 - 16x_{24}x_{27} - 4x_{27}^2 \end{aligned}$$

$$\begin{aligned} R_{28} = & -1 + 8x_9 + 8x_{23} + x_{43} - 4x_9^2 \\ & - 8x_9x_{23} - 4x_{23}^2 \end{aligned}$$

$$\begin{aligned} R_{29} = & -1 + x_{44} - 16x_9^2 - 8x_9x_{10} - 32x_9x_{23} \\ & - 16x_9x_{43} - x_{10}^2 - 8x_{10}x_{23} - 4x_{10}x_{43} - 16x_{23}^2 \\ & - 16x_{23}x_{43} - 4x_{43}^2 \end{aligned}$$

$$\begin{aligned} R_{30} = & 3 + x_9 - x_{23} + x_{45} - 2x_9x_{10} \\ & - 4x_9x_{44} - 3x_{10}^2 - 8x_{10}x_{44} - 4x_{44}^2 \end{aligned}$$

$$R_{31} = -1 - 4x_{45} + x_{46} - 4x_{45}^2$$

$$\begin{aligned} R_{32} = & -1 + x_{47} - x_{10}^2 - 8x_{10}x_{45} - 4x_{10}x_{46} \\ & - 16x_{45}^2 - 16x_{45}x_{46} - 4x_{46}^2 \end{aligned}$$

$$\begin{aligned} R_{33} = & 2 + x_9 + x_{48} - x_9^2 - 4x_9x_{10} \\ & - 4x_9x_{47} - 4x_{10}^2 - 8x_{10}x_{47} - 4x_{47}^2 \end{aligned}$$

$$R_{34} = 1 + x_{49} - 4x_{42}^2 - 8x_{42}x_{48} - 4x_{48}^2$$

$$\begin{aligned} R_{35} = & 1 + x_8 + x_{50} - x_8^2 - 8x_8x_{42} \\ & - 4x_8x_{49} - 16x_{42}^2 - 16x_{42}x_{49} - 4x_{49}^2 \end{aligned}$$

$$R_{36} = -1 - 4x_{50} + x_{51} - 4x_{50}^2$$

$$\begin{aligned} R_{37} = & -1 + x_{52} - x_{10}^2 - 8x_{10}x_{50} - 4x_{10}x_{51} \\ & - 16x_{50}^2 - 16x_{50}x_{51} - 4x_{51}^2 \end{aligned}$$

$$R_{38} = 2 + x_8 + x_{53} - x_8^2 - 4x_8x_{10} - 4x_8x_{52} - 4x_{10}^2 - 8x_{10}x_{52} - 4x_{52}^2$$

$$R_{39} = -10 - 12x_{53} + x_{54} - 4x_{53}^2$$

$$\begin{aligned} R_{40} = & -19 + 4x_9 + 4x_{23} - 16x_{53} + 6x_{54} \\ & + x_{55} - 4x_9^2 - 8x_9x_{23} - 16x_9x_{53} - 8x_9x_{54} \\ & - 4x_{23}^2 - 16x_{23}x_{53} - 8x_{23}x_{54} - 24x_{53}^2 - 16x_{53}x_{54} \\ & - 4x_{54}^2 \end{aligned}$$

$$\begin{aligned} R_{41} = & -10 + x_9 - 12x_{48} + x_{56} - 25x_9^2 \\ & - 40x_9x_{23} - 20x_9x_{55} - 16x_{23}^2 - 16x_{23}x_{55} - 4x_{48}^2 \\ & - 4x_{55}^2 \end{aligned}$$

$$R_{42} = 21 + 24x_{48} - 2x_{56} + x_{57} - 8x_{48}^2 - 16x_{48}x_{56} - 4x_{56}^2$$

$$\begin{aligned} R_{43} = & -1 + 4x_8 + 8x_9 + x_{10} - 2x_{19} \\ & + x_{58} + 12x_{59} - x_{10}^2 - 4x_{10}x_{57} - 4x_{57}^2 \\ & - 4x_{59}^2 \end{aligned}$$

$$\begin{aligned} R_{44} = & -22 - 38x_8 - 76x_9 + 11x_{10} + 19x_{19} \\ & + x_{59} - 16x_8^2 - 64x_8x_9 + 8x_8x_{10} + 16x_8x_{19} \\ & - 64x_9^2 + 16x_9x_{10} + 32x_9x_{19} - x_{10}^2 - 4x_{10}x_{19} \\ & - 4x_{19}^2 \end{aligned}$$

$$\begin{aligned} R_{45} = & -4 + 8x_8 + 16x_9 - 4x_{19} + 24x_{59} \\ & + x_{60} - 4x_{58}^2 - 8x_{59}^2 \end{aligned}$$

$$R_{46} = -1 - 4x_{60} + x_{61} - 4x_{60}^2$$

$$R_{47} = x_{62} - x_2x_{13}$$

$$R_{48} = x_{12} - x_{13} - x_{63} - 2x_{22}x_{62}$$

$$R_{49} = x_{12} - x_{13} - x_{64} - 2x_{26}x_{62}$$

$$R_{50} = x_{12} - x_{13} - 2x_{62} - x_{65} - 2x_{28}x_{62}$$

$$R_{51} = x_{12} - x_{13} - x_{66} - 2x_{30}x_{62}$$

$$R_{52} = x_{12} - x_{13} + 2x_{62} - x_{67} - 2x_{32}x_{62}$$

$$R_{53} = x_{12} - x_{13} - x_{68} - 4x_{34}x_{62} - 2x_{35}x_{62}$$

$$R_{54} = x_{12} - x_{13} - x_{69} - 2x_{38}x_{62}$$

$$R_{55} = -2x_{62} + x_{65} - x_{70} + 2x_{28}x_{62} - 2x_{41}x_{62}$$

$$R_{56} = x_{71} - x_8x_9 + x_8x_{10} + x_9x_{10} - x_{10}^2$$

$$R_{57} = 1 + 2x_{11} + x_{72} + x_{11}^2 - x_{71}^2$$

$$R_{58} = x_{73} - x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{74} - x_{13}^2 - 2x_{13}x_{74} - x_{72}^2 - x_{74}^2$$

$$R_{59} = x_{74} - 4x_{60}x_{62} - 2x_{61}x_{62}$$

$$R_{60} = -x_{13} + x_{77} - x_9x_{62} + x_{75}x_{76}$$

$$R_{61} = -1 + 8x_2 + x_{75} + 4x_2x_8 + 8x_2x_9 - 2x_2x_{19}$$

$$R_{62} = -x_{12} + x_{76} - x_8x_{201}$$

$$R_{63} = x_{67} + x_{78} + x_9x_{201} - 2x_{19}x_{62} + 2x_{32}x_{62}$$

$$R_{64} = x_{79} - x_{12}x_{63} + x_{13}x_{63}$$

$$R_{65} = x_{80} - x_{64}x_{79}$$

$$R_{66} = x_{81} - x_{65}x_{80}$$

$$R_{67} = x_{82} - x_{66}x_{81}$$

$$R_{68} = x_{83} - x_{67}x_{82}$$

$$R_{69} = x_{84} - x_{68}x_{83}$$

$$R_{70} = x_{85} - x_{69}x_{84}$$

$$R_{71} = x_{86} - x_{70}x_{85}$$

$$R_{72} = x_{87} - x_{77}x_{86}$$

$$R_{73} = x_{88} - x_{78}x_{87}$$

$$R_{74} = x_{89} - x_{88}^2$$

$$R_{75} = -x_4 + 2x_{16} + x_{90} - 2x_4x_{16}$$

$$R_{76} = x_{91} - x_{14}x_{90}$$

$$\begin{aligned} R_{77} = & -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} \\ & - x_{12}^2 - 2x_{12}x_{91} - x_{13}^2 - 2x_{13}x_{92} + x_{17}x_{93} \\ & - x_{73}x_{89} - x_{91}^2 - x_{92}^2 \end{aligned}$$

$$R_{78} = x_{15} + x_{92} - x_4x_{15} - x_{15}x_{90}$$

$$R_{79} = x_{97} + x_{99} - x_{94}x_{97}$$

$$R_{80} = -x_{98} - x_3x_{120} + x_{95}x_{193} - x_{99}x_{120}$$

$$R_{81} = -x_{102} + x_{100}x_{193} - x_{101}x_{123}$$

$$R_{82} = -x_{105} + x_{103}x_{193} - x_{104}x_{123}$$

$$R_{83} = -x_{108} + x_{106}x_{193} - x_{107}x_{123}$$

$$R_{84} = -x_{111} + x_{109}x_{193} - x_{110}x_{123}$$

$$R_{85} = -x_{115} + x_{112}x_{194} - x_{114}x_{123}$$

$$R_{86} = -x_{118} + x_{116}x_{194} - x_{117}x_{123}$$

$$R_{87} = x_{123} - x_{17}x_{120}$$

$$R_{88} = x_{119} - x_{122} - x_{121}x_{123}$$

$$R_{89} = x_{124} - x_{126} - x_{123}x_{125}$$

$$R_{90} = -x_{129} - x_{123}x_{128} + x_{127}x_{193}$$

$$R_{91} = x_{130} + x_{133} - 2x_{130}x_{132}$$

$$R_{92} = -x_{133} + x_{134} - x_{130}x_{133}$$

$$R_{93} = -1 + x_{131}^2 + x_{133}^2 - x_{134}^2$$

$$R_{94} = 1 - x_{135} + x_{137} - x_{131}x_{136}$$

$$R_{95} = -1 + 2x_{135} - x_{135}^2 + 2x_{135}x_{137} + x_{136}^2$$

$$R_{96} = 1 + x_{138} + x_3^2 - 2x_3x_{131}$$

$$R_{97} = x_{17} - x_{169} - x_3x_{170} + x_{138}x_{139}$$

$$\begin{aligned} R_{98} = {} & -5 - 8x_2 - 16x_{16} + 4x_{131} - x_{140} \\ & - 4x_2^2 - 16x_2x_{16} + 4x_2x_{131} - 16x_{16}^2 + 8x_{16}x_{131} \end{aligned}$$

$$\begin{aligned} R_{99} = {} & -1 + x_{94} - x_{169} - 2x_{170} - 2x_2x_{170} \\ & - 4x_{16}x_{170} + x_{140}x_{141} \end{aligned}$$

$$R_{100} = -x_{142} + x_{144} - x_{131}x_{143}$$

$$R_{101} = -1 - x_{142}^2 + 2x_{142}x_{144} + x_{143}^2$$

$$R_{102} = -x_{145} + x_{147} - x_{131}x_{146}$$

$$R_{103} = -1 - x_{145}^2 + 2x_{145}x_{147} + x_{146}^2$$

$$R_{104} = -x_{148} + x_{150} - x_{131}x_{149}$$

$$R_{105} = -1 - x_{148}^2 + 2x_{148}x_{150} + x_{149}^2$$

$$R_{106} = -x_{151} + x_{153} - x_{131}x_{152}$$

$$R_{107} = -1 - x_{151}^2 + 2x_{151}x_{153} + x_{152}^2$$

$$R_{108} = -x_{154} + x_{156} - x_{131}x_{155}$$

$$R_{109} = -1 - x_{154}^2 + 2x_{154}x_{156} + x_{155}^2$$

$$R_{110} = -x_{157} + x_{159} - x_{131}x_{158}$$

$$R_{111} = -1 - x_{157}^2 + 2x_{157}x_{159} + x_{158}^2$$

$$R_{112} = 1 - x_{160} + x_{162} - x_{131}x_{161}$$

$$R_{113} = -1 + 2x_{160} - x_{160}^2 + 2x_{160}x_{162} + x_{161}^2$$

$$R_{114} = -x_{163} + x_{165} - x_{131}x_{164}$$

$$R_{115} = -1 - x_{163}^2 + 2x_{163}x_{165} + x_{164}^2$$

$$R_{116} = -x_{166} + x_{168} - x_{131}x_{167}$$

$$R_{117} = -1 - x_{166}^2 + 2x_{166}x_{168} + x_{167}^2$$

$$R_{118} = 1 - x_{169} + x_{171} - x_{131}x_{170}$$

$$R_{119} = -1 + 2x_{169} - x_{169}^2 + 2x_{169}x_{171} + x_{170}^2$$

$$R_{120} = -x_{172} + x_{174} - x_{131}x_{173}$$

$$R_{121} = -1 - x_{172}^2 + 2x_{172}x_{174} + x_{173}^2$$

$$R_{122} = -x_{175} + x_{177} - x_{131}x_{176}$$

$$R_{123} = -1 - x_{175}^2 + 2x_{175}x_{177} + x_{176}^2$$

$$\begin{aligned} R_{124} = {} & -8 - 2x_0 - 8x_2 - 16x_{16} + 2x_{94} \\ & + x_{130} - 4x_0^2 - 2x_0x_2 - x_1^2 - 6x_2^2 \\ & - 16x_2x_{16} - x_3^2 - x_4^2 - x_8^2 - x_9^2 \\ & - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 \\ & - x_{15}^2 - 20x_{16}^2 - x_{17}^2 - 2x_{94}^2 - x_{95}^2 \\ & - x_{96}^2 - x_{100}^2 - x_{103}^2 - x_{106}^2 - x_{109}^2 \\ & - x_{112}^2 - x_{113}^2 - x_{116}^2 - x_{119}^2 - x_{120}^2 \\ & - x_{124}^2 - x_{127}^2 \end{aligned}$$

$$R_{125} = -32 + x_0 - 128x_{178} - 128x_{178}^2$$

$$R_{126} = x_{189} - x_{131}x_{180}$$

$$R_{127} = -1 + x_{179}^2 + x_{180}^2 - x_{189}^2$$

$$R_{128} = x_{190} - x_{182}x_{183}$$

$$R_{129} = -1 + x_{181}^2 + x_{182}^2 - x_{190}^2$$

$$R_{130} = x_{191} - x_{176}^2$$

$$R_{131} = x_{180} - x_{191} - 2x_{185}x_{191}$$

$$R_{132} = -1 - 4x_{180} + x_{183} - 8x_{180}x_{184}$$

$$R_{133} = -x_{131} + x_{183} - x_{179}x_{186}$$

$$R_{134} = -x_{177} + x_{181} - x_{179}x_{187}$$

$$R_{135} = -2x_{16} + x_{182} - 4x_{176}x_{188}$$

$$\begin{aligned}R_{136} = 9 &- 2x_{144} - 2x_{147} - 2x_{150} - 2x_{153}\\ &- 2x_{156} - 2x_{159} - 2x_{165} - 2x_{168} - 2x_{174}\\ &- 2x_{177} + x_{192} - x_{197} + 2x_0^2 + 2x_0x_2\\ &- 2x_0x_{136} - 2x_0x_{143} + x_1^2 - 2x_1x_{170} + x_2^2\\ &- 2x_2x_{143} + x_4^2 - 2x_4x_{161} + x_8^2 - 2x_8x_{146}\\ &+ x_9^2 - 2x_9x_{149} + x_{10}^2 - 2x_{10}x_{152} + x_{11}^2\\ &- 2x_{11}x_{155} + x_{12}^2 - 2x_{12}x_{158} + x_{13}^2 - 2x_{13}x_{164}\\ &+ x_{14}^2 - 2x_{14}x_{167} + x_{15}^2 - 2x_{15}x_{173} - 4x_{16}x_{176}\\ &- x_{131}x_{192} + x_{137}^2 + x_{144}^2 + x_{147}^2 + x_{150}^2\\ &+ x_{153}^2 + x_{156}^2 + x_{159}^2 + x_{162}^2 + x_{165}^2\\ &+ x_{168}^2 + x_{171}^2 + x_{174}^2 + 2x_{176}^2 + x_{177}^2\\ &- x_{195}^2 - x_{196}^2 - x_{197}^2\end{aligned}$$

$$R_{137} = x_{120} - x_{96}x_{193}$$

$$R_{138} = x_{120} - x_{113}x_{194}$$

$$\begin{aligned}R_{139} = -1 &- x_{197} - 4x_{16}^2 - x_{136}^2 - x_{143}^2\\ &- x_{146}^2 - x_{149}^2 - x_{152}^2 - x_{155}^2 - x_{158}^2\\ &- x_{161}^2 - x_{164}^2 - x_{167}^2 - x_{170}^2 - x_{173}^2\\ &+ x_{176}^2 - x_{195}^2 - x_{196}^2 - x_{197}^2\end{aligned}$$

$$\begin{aligned}R_{140} = -3 &+ 3x_{200} - 4x_{98}^2 - 4x_{102}^2 - 4x_{105}^2\\ &- 4x_{108}^2 - 4x_{111}^2 - 4x_{115}^2 - 4x_{118}^2 + x_{120}^2\\ &- 4x_{122}^2 - 4x_{126}^2 - 4x_{129}^2 - x_{198}^2 - x_{199}^2\\ &- x_{200}^2\end{aligned}$$

$$R_{141} = x_{201} - x_2x_{12}$$

$$\begin{aligned}R_{142} = x_{202} &+ x_{94}x_{203} - 2x_{95}x_{143} - 2x_{100}x_{146} - 2x_{103}x_{149}\\ &- 2x_{106}x_{152} - 2x_{109}x_{155} - 2x_{112}x_{158} - 2x_{116}x_{164} - 2x_{119}x_{167}\\ &- 2x_{124}x_{173} - 2x_{127}x_{176} - 2x_{131}x_{203} + 2x_{143}x_{144} + 2x_{146}x_{147}\\ &+ 2x_{149}x_{150} + 2x_{152}x_{153} + 2x_{155}x_{156} + 2x_{158}x_{159} + 2x_{164}x_{165}\\ &+ 2x_{167}x_{168} + 2x_{173}x_{174} + 2x_{176}x_{177}\end{aligned}$$

$$\begin{aligned}R_{143} = \; & -10 + x_{203} - x_{94}x_{202} + x_{95}^2 - 2x_{95}x_{142} \\ & + x_{100}^2 - 2x_{100}x_{145} + x_{103}^2 - 2x_{103}x_{148} + x_{106}^2 \\ & - 2x_{106}x_{151} + x_{109}^2 - 2x_{109}x_{154} + x_{112}^2 - 2x_{112}x_{157} \\ & + x_{116}^2 - 2x_{116}x_{163} + x_{119}^2 - 2x_{119}x_{166} + x_{124}^2 \\ & - 2x_{124}x_{172} + x_{127}^2 - 2x_{127}x_{175} + 2x_{142}x_{144} + 2x_{145}x_{147} \\ & + 2x_{148}x_{150} + 2x_{151}x_{153} + 2x_{154}x_{156} + 2x_{157}x_{159} + 2x_{163}x_{165} \\ & + 2x_{166}x_{168} + 2x_{172}x_{174} + 2x_{175}x_{177}\end{aligned}$$

$$\begin{aligned}R_{144} = \; & 2x_{136} + 2x_{161} + 2x_{170} + x_{204} - x_{205} \\ & + x_{94}x_{205} - 2x_{96}x_{136} - 2x_{113}x_{161} - 2x_{120}x_{170} - 2x_{131}x_{205} \\ & + 2x_{136}x_{137} + 2x_{161}x_{162} + 2x_{170}x_{171}\end{aligned}$$

$$\begin{aligned}R_{145} = \; & -2 + 2x_{135} + 2x_{160} + x_{204} + x_{205} \\ & - x_{94}x_{204} + x_{96}^2 - 2x_{96}x_{135} + x_{113}^2 - 2x_{113}x_{160} \\ & + x_{120}^2 - 2x_{120}x_{169} + 2x_{135}x_{137} + 2x_{160}x_{162} + x_{169}^2 \\ & - x_{170}^2\end{aligned}$$

$$R_{146} = -1 + x_4 - x_2x_3$$

$$R_{147} = x_{207} - 524309x_0x_2$$

$$R_{148} = -1 - 2x_2x_{207} + x_{206}^2 - x_{207}^2$$

$$R_{149} = x_0 + x_2 - 524309x_0x_{208}$$

B Artifact map and integration notes

Path	Purpose
`equations/equation.json`	Complete quadratic residual system defining the one quartic.
`equations/expanded_polynomial.json`	All collected integer coefficients of P.
`equations/residuals.txt`	Readable literal residual list.
`equations/variables.tsv`	All 209 coordinate names and source roles.
`equations/transformation.json`	Exact changed rows, parent hash, and index map.
`code/build.py`	Deterministic standard-library producer.
`code/verify.py`	Independent exact coefficient verifier.
`code/check_printed.py`	Parser verifying the printed polynomial.
`code/check_arithmetic.py`	Independent modular arithmetic regression.
`code/evaluate.py`	Exact evaluation on a supplied integer tuple.
`proofs/`	Independent mathematical review notes.
`verification/`	Fresh finite verification receipts, including the parent replay.
`inherited/`	Exact P210 parent archive, its equation bytes, and license.
`PROVENANCE.json`	Snapshot, source identities, and file hashes.
`SHA256SUMS`	Delivered-file integrity manifest.

Suggested additive placement is a new source directory under `Computability/DiophantineConsistency/con-zfc-diophantine-forms/`, with a new source number assigned by the repository maintainer. The archive is suitable for the `docs/incoming/` intake flow. Existing reports and their historical metrics should be preserved. This release has not written to the remote repository.